XLV National Mathematics Olympiad: 4th round, May 1997

Problem 1. Consider the polynomial

$$P_n(x) = \binom{n}{2} + \binom{n}{5}x + \binom{n}{8}x^2 + \cdots + \binom{n}{3k+2}x^k,$$

where $n \geq 2$ is a natural number and $k = \left[\frac{n-2}{3}\right]$.

(a) Prove that $P_{n+3}(x) = 3P_{n+2}(x) - 3P_{n+1}(x) + (x+1)P_n(x)$;

(b) Find all integer numbers a such that $P_n(a^3)$ is divisible by $3^{\left[\frac{n-1}{2}\right]}$ for all $n \geq 2$.

Problem 2. Let M be the centroid of $\triangle ABC$. Prove the inequality

$$\sin \angle CAM + \sin \angle CBM \leq \frac{2}{\sqrt{3}}$$

(a) if the circumscribed circle of $\triangle AMC$ is tangent to the line AB;

(b) for any $\triangle ABC$.

Problem 3. Let n and m be natural numbers such that $m + i = a_i b_i^2$ for $i = 1, 2, \ldots, n$, where a_i and b_i are natural numbers and a_i is not divisible by a square of a prime number. Find all n for which there exists an m such that $a_1 + a_2 + \cdots + a_n = 12$.

Problem 4. Let a, b and c be positive numbers such that $abc = 1$. Prove the inequality

$$\frac{1}{1+a+b} + \frac{1}{1+b+c} + \frac{1}{1+c+a} \leq \frac{1}{2+a} + \frac{1}{2+b} + \frac{1}{2+c}.$$

Problem 5. Given a $\triangle ABC$ with bisectors BM and CN ($M \in AC$, $N \in AB$). The ray $MN^{\rightarrow}$ intersects the circumcircle of $\triangle ABC$ at point D. Prove that

$$\frac{1}{BD} = \frac{1}{AD} + \frac{1}{CD}.$$

Problem 6. Let X be a set of $n+1$ elements, $n \geq 2$. Ordered n-tuples $(a_1, a_2, \ldots, a_n)$ and $(b_1, b_2, \ldots, b_n)$ formed from distinct elements of X are called 'disjoint' if distinct indices i and j exist such that $a_i = b_j$. Find the maximal number of n-tuples any two of which are 'disjoint'.

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